Computational Fabrication by Multi-Axis 3D Printing

Foundations and Hands-on Practice

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Figure placeholder. A teaser concept image showing the whole course idea: an input shape and design objectives leading to curved layers, tool orientations, non-planar toolpaths, machine-aware execution, and printed results on robot, five-axis, and three-axis platforms.

Figure 1: Teaser concept for computational design and fabrication by multi-axis 3D printing.

Abstract

Multi-axis 3D printing changes additive manufacturing from a layer-stacking process into a coupled computational problem involving curved layers, tool orientations, path planning, support, collision, material anisotropy, surface quality, and machine constraints. This course presents multi-axis printing as a computational design-to-fabrication framework with two connected threads. The first thread surveys the state of the art in multi-axis algorithms and applications, focusing on how representations, constraints, and fabrication objectives are formulated and verified. The second thread asks how these ideas can be transferred to ordinary desktop three-axis FFF printers, where the nozzle orientation is fixed but non-planar surface-following toolpaths can improve surface quality, reduce stair-stepping, partially reduce support, and provide limited path-driven material-orientation control. The course emphasizes both the frontier and the practical boundary: what requires robotic or five-axis hardware, and what can already be explored on accessible desktop machines. **[Charlie:: to update after discussing the course content]**

1 Course Presenters

- **Charlie C.L. Wang** is the Professor and Chair in Smart Manufacturing, Director of Digital Manufacturing Lab, and Department Head of Research with The University of Manchester. His research spans geometric computing, computational design, additive manufacturing, robot-assisted fabrication, and support-aware process planning. His work has transformed the integration of design, analysis, and fabrication by establishing geometric computing and optimization as foundations for intelligent engineering across additive and hybrid manufacturing.
- **Chengkai Dai** is a Postdoctoral Fellow in the Personalized Design and Fabrication Lab at the Centre for Perceptual and Interactive Intelligence, The Chinese University of Hong Kong, working with Professor Yeung Yam and previously supervised by Professor Charlie C.L. Wang. His research focuses on computational fabrication, additive manufacturing, and textile-based fabrication, developing computational and robotic methods that bridge digital design and physical fabrication. His work includes multi-axis slicing and robotic motion planning for freeform additive manufacturing, as well as knitting and weaving algorithms for functional-fiber sensing interfaces and robotic skins.
- **Guoxin Fang** leads the Computational Robotics and Manufacturing Lab in the Department of Mechanical and Automation Engineering at The Chinese University of Hong Kong. His group explores the intersection of computational science and engineering innovation through geometry computing, physical modeling, and artificial intelligence, developing technologies for robotics, advanced manufacturing, and smart systems. His research interests include robotic multi-axis spatial additive manufacturing, modeling and perception for soft robots, proprioception, and computational design and digital manufacturing for personalized multifunctional wearable devices.
- **Neelotpal Dutta** is a Postdoctoral Fellow in the Department of Mechanical and Aerospace Engineering at The University of Manchester, supervised by Professor Charlie C.L. Wang. His research interests include geometric computing, digital manufacturing, and computer-aided design and manufacturing, with a focus on toolpath generation and process planning for additive and subtractive manufacturing.
- **Tao Liu** is a Ph.D. student at The University of Manchester, supervised by Professor Charlie C.L. Wang, working on neural-based design and manufacturing optimization. His research focuses on computational geometry, additive manufacturing, and robotics, with particular interests in multi-axis 3D printing, topology optimization, neural implicit representations for geometric modeling, surface reconstruction, fiber-reinforced composites, and material optimization.
- **Yongxue Chen** is a Ph.D. student in robot-assisted manufacturing with the Digital Manufacturing Laboratory, School of Engineering, The University of Manchester, supervised by Professor Charlie C.L. Wang. His research interests include robotics and additive manufacturing.

[Charlie:: more speaker to add? suggested: Tao, Neelotpal, Guoxin and Chengkai?] **[Tao:: The max number of lecture is 5 and suggested 2-3 lectures]**

2 Course Format

The course is planned as a tutorial-style session with visual examples, representative papers, and compact hands-on planning exercises. The format separates foundation topics from practice:

Part One builds the planning vocabulary, and Part Two turns selected ideas into five-axis and three-axis workflow examples.

2.1 Schedule

[Charlie:: 1: Introduction (Charlie); 2: Multi-axis algorithm from scale-field (SIG 2018 - Chengkai Dai); 3: Neural Field + S^3 (Neelotpal + Tao); 4: Anisotropic (SIGA 2020 + AM - Guoxin Fang); 5: Co-Optimization (Tao); 6: Collision-free (Neelotpal + Guoxin) 7: IK + Support Generation (Yongxue + Neelotpal); 8: Practice and Code (All) 9: Conclusion (Charlie)]

The course is organized as a 3 hour 45 minute session, including a 15 minute break. The lecture portion takes 1 hour 55 minutes, and the hands-on practice and discussion take 1 hour 35 minutes.

2.1.1 Part One: Background and Foundations (1 hour 55 minutes).

- **Introduction and motivation.(10 minutes)** **[Professor]**Why multi-axis and non-planar printing change the design-to-fabrication problem, and how the course connects robot printing, five-axis planning, and accessible three-axis workflows.
- **Scalar fields and support-free planning.** **[Chengkai]**Curved layers, iso-values of scalar fields, support-aware decomposition, and representative field-based formulations for multi-axis printing [Dai et al. 2018; Wu et al. 2020].
- **Reinforced FDM and anisotropic fiber-aware deposition.** **[Guoxing]**How path direction, filament alignment, material orientation, and reinforcement goals influence slicing and toolpath design [Fang et al. 2024, 2020; Zhang et al. 2025].
- **S^3-Slicer, Neural Slicer, and structure–fabrication co-optimization.****[Tao]** Deformation-based slicing, rotation and scale fields, ARAP-style regularization, layer-generation maps, neural slicing-field optimization, and mechanically informed fabrication constraints [Liu et al. 2024, 2025; Zhang et al. 2022].
- **Collision-free planning and toolpath generation.** **[Neel]**Accessibility, clearance, collision checking, continuous toolpaths, inverse-kinematics continuity, and support-aware execution [Dutta et al. 2025; Qu et al. 2025].

2.1.2 Break (15 minutes).

2.1.3 Part Two: Planning Concepts Hands-On and Discussion (1 hour 35 minutes).

(1) **Five-axis planning, toolpaths, and IK**
   - Introduce collision-aware planning using implicit neural fields and INF-3DP.
   - Compare collision-aware planning with semi-continuous toolpath generation in AtomSlicer.
   - Discuss IK refinement through the co-optimization of tool orientations, kinematic redundancy, and waypoint timing.
   - Instructor explains one small planning example, followed by a short participant task.

(2) **Three-axis planning**
   - Introduce fixed-nozzle constraints for manufacturable layers, path orientations, and reinforcement planning.
   - Compare reinforcement objectives with surface-quality objectives on desktop FFF hardware.
   - Attendees solve a small planning or parameter-selection task.

(3) **From planning to printing**
   - Connect field planning, support generation, slicing, toolpath generation, G-code export, and hardware handoff.
   - Discuss practical checks that decide whether a curved-layer plan can become executable motion and a reliable print.
   - Explain optional follow-up practice materials.

(4) **Discussion and wrap-up**
   - Discuss open problems in planning robustness, verification, material-aware slicing, machine constraints, and software accessibility.
   - Connect the lecture topics and hands-on examples to possible follow-up projects.

Detailed practice workflows are reserved for the Part Two notes and accompanying materials, so the schedule here focuses on the sequence, scope, and timing of the course.

2.2 Audience and prerequisites

The target audience is students, researchers, and practitioners in computer graphics, computational fabrication, digital design, additive manufacturing, and robotics. Basic familiarity with triangle meshes, surface normals, slicing, toolpaths, and material-extrusion 3D printing is helpful. Experience with robot programming, inverse kinematics, finite-element simulation, or implicit neural representations is useful but not required.

3 Description of Course

This course introduces computational fabrication methods for multi-axis and non-planar 3D printing, with emphasis on how geometric and physical objectives become manufacturable layers, toolpaths, and executable machine motion. The course is organized around two connected parts. Part One builds the technical background through representative planning concepts, while Part Two turns selected concepts into compact hands-on workflows for five-axis planning, inverse kinematics, three-axis constrained planning, and planning-to-printing practice.

3.1 Part One: background and foundations

Part One follows the main decisions made by a fabrication planner rather than presenting the literature as isolated examples. It begins with layer representations and support-aware planning, then asks how material direction and reinforcement change slicing and toolpath design. The discussion then moves to deformation-based and neural slicing fields, structure–fabrication co-optimization, and finally the machine-aware constraints that determine whether a planned path can be executed: accessibility, collision avoidance, toolpath continuity, inverse kinematics, and support generation.

3.1.1 Introduction and motivation. **[Professor]**

The opening topic frames multi-axis and non-planar printing as a planning problem. A conventional slicer primarily selects horizontal layers and in-layer paths such as contours, zig-zag infill, or parallel rasters. Once layers become curved, deposition directions vary, or material orientation becomes part of the design objective, the planner must also reason about support, anisotropy, collision, and execution feasibility. This framing prepares attendees to read the following topics as different ways of representing, optimizing, and validating a fabrication plan.

3.1.2 Scalar fields and support-free planning. **[Chengkai]**
The first technical topic introduces scalar fields as a common representation for curved-layer planning. Iso-values of a scalar field define layers, while the field structure can guide toolpaths, layer thickness, surface following, and support-aware deposition. Support-free volume printing is used as the representative field-based example for multi-axis support-aware deposition [Dai et al. 2018]. General support-effective decomposition is used as the discrete counterpart, where a volume is decomposed into manufacturable regions and printing directions before toolpath generation [Wu et al. 2020].

3.1.3 Reinforced FDM and anisotropic fiber-aware deposition. **[Guoxin]**
The next topic turns from where to place layers to what the deposited paths should achieve. In anisotropic and fiber-reinforced extrusion, path direction is not only a geometric choice but also a mechanical design variable. Reinforced FDM is used as the first representative example, showing how multi-axis filament alignment can be planned to control anisotropic strength [Fang et al. 2020]. Spatial printing with continuous fiber extends this idea to curved slicing, continuous-fiber toolpath generation, and physical verification in an additive-manufacturing workflow [Fang et al. 2024]. Stress-guided spatial fiber printing then shows how principal stresses can guide high-density fiber toolpaths for composite fabrication [Zhang et al. 2025]. Together, these papers motivate material-aware planning as a coupling between geometry, path direction, reinforcement layout, and mechanical performance.

3.1.4 S^3-Slicer, Neural Slicer, and structure–fabrication co-optimization. **[Tao]**
The course then studies how curved layers can be generated through field and mapping formulations. S^3-Slicer is introduced as the representative deformation-based slicing framework for multi-axis 3D printing [Zhang et al. 2022]: slicing is formulated through a map from the input volume to a printing-aware parameterization, leading to rotation and scale fields, ARAP-style regularization, and a layer-generation map whose iso-values define curved layers. Neural Slicer then shows how this type of slicing field can be represented and optimized by a neural model [Liu et al. 2024]. Neural co-optimization extends the same computational view from layer generation to coupled structural topology, manufacturable layers, and path orientations for fiber-reinforced composites [Liu et al. 2025].

3.1.5 Collision-free planning and toolpath generation. **[Neel]**
The final Part One topic moves from layer and path design to execution constraints. A visually plausible curved-layer plan is not enough: the tool must remain accessible, avoid collision, and satisfy clearance and layer-thickness constraints. INF-3DP is used as the collision-free multi-axis printing example based on implicit neural fields [Qu et al. 2025], while implicit neural field-based process planning is used to discuss direct control over both collision avoidance and toolpath geometry [Dutta et al. 2025]. The topic then connects collision-free planning to executable machine motion through semi-continuous field-aligned toolpaths, singularity-aware motion planning, IK continuity, waypoint timing, and support generation [Chen et al. 2025; Cocco et al. 2026; Zhang et al. 2021, 2023]. This topic sets up the hands-on part by showing where planning ends and machine-aware verification begins.

Figure placeholder. A Part One overview image comparing core representations and planning concepts: scalar fields, iso-value layers, anisotropic fiber-aware deposition, neural slicing fields, structure–fabrication co-optimization, collision checking, toolpath generation, inverse kinematics, and support generation.

Figure 2: Part One overview: representations and planning foundations.

3.2 Part Two: hands-on planning and discussion

The second part connects the lecture topics to executable planning workflows. It is organized around three practice modules followed by discussion and conclusion. The goal is to help attendees translate representation choices into concrete planning decisions: what is optimized, what constraints are enforced, and what must be checked before printing.

3.2.1 Five-axis planning and inverse kinematics. **[Neel, Tao]**
The five-axis planning module compares collision-aware and unconstrained toolpath planning. Implicit neural field-based process planning is used to discuss direct control over collision avoidance and toolpath geometry [Dutta et al. 2025]. INF-3DP is used as the collision-free multi-axis 3D printing example [Qu et al. 2025]. AtomSlicer is used to discuss a semi-continuous formulation in which toolpaths can remain mostly continuous while still allowing discontinuities or singular behavior when the fabrication plan requires them [Cocco et al. 2026]. The IK component then introduces the co-optimization of tool orientations, kinematic redundancy, and waypoint timing for robot-assisted manufacturing [Chen et al. 2025].

3.2.2 Three-axis constrained planning. **[Neel, Tao]**
The three-axis module explains how multi-axis planning ideas change under fixed-nozzle constraints. Ordinary desktop FFF printers cannot rotate the tool, but they can still use non-planar surface-following toolpaths to improve top-surface quality, reduce stair-stepping, partially reduce support, and provide limited path-driven material-orientation control. Neural co-optimization for fiber-reinforced composites is used to discuss fixed-nozzle constraints on manufacturable layers, path orientation, and reinforcement planning [Liu et al. 2025]. AtomSlicer's FFF-focused non-planar slicing example is used when comparing reinforcement objectives with surface-quality objectives [Cocco et al. 2026].

Figure placeholder. A Part Two workflow image showing hands-on examples: five-axis planning with collision-aware and unconstrained toolpaths, IK refinement, three-axis fixed-nozzle planning, support generation, slicing, toolpath generation, G-code export, and Bambu printing.

Figure 3: Part Two overview: hands-on planning workflows.

3.2.3 From planning to printing. The practice workflow connects field planning, support generation, slicing, toolpath generation, G-code export, and hardware handoff. For reinforcement-oriented examples, the workflow starts from an input mesh and boundary conditions, uses FEA or stress information to align paths and reinforcement directions, then proceeds through printing-field planning, support generation, slicing, toolpath generation, G-code generation, and Bambu printing. For surface-quality examples, the workflow keeps the same manufacturability constraints but replaces the reinforcement objective with surface-following and surface-quality criteria.

3.2.4 Discussion and conclusion. **[Professor]**

The discussion returns to the practical questions that determine whether a multi-axis or non-planar plan can move from geometry to fabrication. Attendees compare collision-free and unconstrained planning, discrete decomposition, continuous scalar-field planning, and semi-continuous formulations. The course also summarizes open problems in verification, software accessibility, material-aware slicing, IK continuity, singularity reduction, support-aware planning, and reliable robot printing. The intended outcome is a practical way to read new papers, evaluate fabrication plans, and identify which constraints are guaranteed by a planner and which still need to be checked before printing.

4 Learning Objectives and Outcomes

This course is intended to build a coherent view of multi-axis and non-planar 3D printing as a computational design-to-fabrication problem. It traces how the field has moved beyond planar slicing and support generation toward curved layer design, field-based slicing, collision-aware toolpath planning, and structure–fabrication co-optimization.

The course emphasizes how to reason through the complete pipeline from geometry and material objectives to curved layers, executable toolpaths, manufacturability constraints, and machine-specific execution. Along the way, participants are introduced to the main computational representations used in this area, including scalar fields, deformation fields, frame fields, phase fields, voxel and mesh-based methods, graph planners, and neural fields. The goal is to make these methods comparable: what each representation makes easy, what constraints it captures, and what still needs to be checked before fabrication.

A further objective is to clarify the boundary between different hardware platforms. Ordinary three-axis non-planar printing, desktop rotary or five-axis systems, and robot-assisted fabrication expose different levels of tool-pose freedom and different practical constraints. By the end of the course, attendees should be able to judge which multi-axis ideas can be transferred to accessible desktop material-extrusion printers, which require richer machine motion, and how this broader computational framework can inform future software tools for non-planar and reinforcement-aware fabrication.

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